

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Electronics and Electrical Communication Engineering
2. **Subject Name:** Communication-II Laboratory. **L-T-P:** 0-0-3 **Credits:**2
3. **Pre-requisites:** Communication-I Laboratory,
4. **Syllabus and reference books:**

Syllabus: Generation of PN sequence with specific properties of autocorrelation and cross correlation. Design of BPSK and QPSK modulator and demodulators. Clock recovery of digital signals. Exploring more base-band modulation and demodulation techniques. Channel estimation and channel equalizers. Source coding and channel coding such as Huffman coding, hamming codes and convolutional codes. Viterbi decoding. Advanced wireless communication experiments related to fading counter measures such as OFDM and space time block codes. This lab is a balanced mixture of hardware and software experiments.

Reference Books:

- 1) Gallager RG. *Principles of Digital Communication*. Cambridge University Press; 2008.
- 2) Proakis, J. G., Salehi, M. (2008). *Digital Communications*. United States: McGraw-Hill.
- 3) Goldsmith, A. (2005). *Wireless Communications*. Senegal: Cambridge University Press.
- 4) Tse, D., Viswanath, P. (2005). *Fundamentals of wireless communication*. United Kingdom: Cambridge University Press.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Generation of advanced PN sequences	6=2*3
2.	BPSK modulation and demodulation	3=1*3
3.	QPSK modulation and demodulation	3=1*3
4.	Clock recovery	6=2*3
5.	Base-band modulations using software/ SDR	6=2*3
6.	Source coding software/SDR	6=1*3
7.	Channel coding- decoding software/SDR	6=2*3
8.	Channel estimation and equalization	6=2*3
Total number of hours		42 lab hours